

Can Qwen vision replace OCR and Python?

A measured answer from prior Coulton Creek runs. These are development benchmarks, not a production accuracy claim.

Short answer: No, not as an unguarded single extractor in the tested setup. Qwen is valuable for visual meaning; native text/OCR preserve literal evidence, and geometry plus validation preserve structure.

1. Same-page, same-model ablation: dense financial table (page 6)

Qwen3.5:9B used the same page, prompt, schema, decoding settings and frozen 61-fact reference; only the supplied image/OCR input changed. "OCR-only" here means Qwen reading OCR text, not a RapidOCR/Python-only pipeline.

Qwen input	Exact complete facts	Other useful signal	Wall time
Image only	0 / 61	95.1% evidence recall; 23 unsupported/duplicate facts	20.4 min
OCR text only	0 / 61	100% evidence recall; 16 unsupported/duplicate facts	22.8 min
OCR + image	7 / 61 (11.5% recall; 17.5% F1)	3 unsupported/duplicate facts	9.6 min

A fact counted only when *all* owner, page/section/row, relation, printed value, period, modality, polarity, condition and evidence fields matched. Thus 0/61 does **not** mean the model read zero characters; it means no entire fact record passed this strict test. The reference was assistant-verified, not independently human-audited.

2. Cover-page legal detail (page 1)

Method	Semantic items	Observed failure
Qwen3.5:4B, page image only	6 / 12	Inverted the "no money solicited" restriction; omitted legal details.
Qwen3.5:4B, image + native + OCR	8 / 12	Valid JSON, but still omitted four reference items.
Native text / independent OCR	12 / 12 each	Literal semantics available, but neither captured the brand graphic role.

The 4B cover test and 9B table test are different models/tasks; their fractions must not be pooled.

3. What OCR alone actually measures

Fresh 16-page, 200-DPI RapidOCR 3.9.2 + PP-OCRv6 bake-off: **94.68% macro token F1, 93.15% numeric F1, 89.03% numeric recall**, 0 failed pages, **2.17 sec/page**. It read page-6 table text at about 99% token F1, yet emitted text lines—not a trustworthy row/column grid. On a separate 31-page / 7-PDF stress set: **87.59% token F1, 85.85% numeric F1**, 100% normal completion, **3.11 sec/page**.

OCR transcription F1 and source-block/fact F1 are different endpoints; there is no directly comparable "Qwen versus RapidOCR accuracy" number here.

Why a single-stage approach breaks

Failures observed in the recorded trials, and the engineering implication for the current pipeline.

Observed failure	Why it matters	Guardrail
Literal text changes or is compressed	A negated legal restriction was inverted on page 1; omitted clauses alter meaning.	Keep native/OCR text and require evidence-grounded checks for negation and conditions.
Table values lose their cell/section owner	Page 6 had high evidence coverage but near-zero exact complete facts in direct ablations.	Use PDF/table geometry; reconstruct cells and section scope deterministically; verify printed tokens.
JSON can parse without being faithful	The page-1 hybrid produced valid JSON but only 8/12 required semantic items.	Validate both schema and source completeness/grounding; do not equate "valid JSON" with correct extraction.
Latency and long outputs	The 9B page-6 direct calls took 9.6-22.8 minutes on the recorded local runtime.	Keep VLM tasks narrow and bounded; stop repetition/truncation; reserve models for semantic linking.

The result is a **division of labor**: literal acquisition (native text + RapidOCR), structural recovery (PDF geometry / Python / OpenCV), mandatory Qwen visual-semantic linking for charts and maps, then deterministic reconstruction and source-backed validation.

What the historical tests do—and do not—prove

They prove: image-only Qwen3.5:4B missed half the cover's semantic items; direct Qwen3.5:9B table-input variants failed strict complete-fact scoring; high OCR transcription does not itself produce a reliable source-block hierarchy.

They do not prove: that every Qwen model fails, that RapidOCR/Python alone can solve charts/maps, or that a current Qwen3-VL-only run has been scored on the same 16-page rubric. The recent Page 7 10/10 pie observations came from the *combined* Qwen3-VL + OCR/native + PDF/OpenCV/Python pipeline, not Qwen alone.

A post-review Page 6 OCR structural-rules experiment reached 61/61 on that one page. Because its rules were developed after inspecting Page 6 and the OCR already contained a clean HTML table, this is a development result—not unseen-document generalization.

Recommendation / next valid benchmark

Keep all three structured stages. For Rahul's proposed replacement, freeze one blind, multi-document test set and score **current Qwen3-VL image-only**, **native+RapidOCR+geometry without VLM**, and **full hybrid** on the same pages, same source-block reference, exact cell/label bindings, completeness, unsupported facts, schema validity, and runtime. Only that is a head-to-head architecture decision.

Evidence and reproducibility

Original PDF SHA-256: cf569a8b25fcd36301b34e0e976e88227faa9f9cdc7804f34eddad09e8d477ad. Local recorded artifacts: document-pipeline/page-benchmarks/coulton-page-001.md; pipeline-runs/one-page-p006-evaluation-alternative-methods-20260904-r2/COMPARISON.md and variant manifests; experiments/ocr-bakeoff-from-scratch-20260908/REPORT.md; experiments/ocr-stratified-bakeoff-from-scratch-20260908/REPORT.md. These historical raw artifacts are outside this GitHub repository.